

One Evidence Standard Now. One Exact Benchmark Next.

Giovanni Della-Libera

Internal Candidate Policy Brief — July 2026

The proposal. Require every State to publish a reproducible geographic benchmark, its final congressional plan, every changed block, alternatives, effects, public comments, and reasons.

The North Star. Replace the heuristic benchmark with an exact, uniquely selected plan backed by a mathematical proof certificate.

What We Can Defend Today

BISECT can produce and verify a partisan-input-excluded geographic benchmark. It records code, inputs, parameters, assignments, analysis, and reports.

That claim is useful but limited:

- it proves execution, not fairness;
- it does not establish Voting Rights Act compliance;
- it does not preserve every community;
- it does not prove a global optimum; and
- it does not replace a commission, court, or public process.

The model federal bill therefore treats the benchmark as mandatory *evidence*, not the mandatory final map.

Why This Is Still A Major Reform

Today, a final map can hide thousands of choices. Under the proposed standard, the public receives:

1. one precommitted benchmark;
2. one legally adopted final plan;
3. a block-by-block diff;
4. legal authority for every change;
5. a lower-departure lawful alternative for discretionary changes;
6. public and independent review; and
7. a permanent challenge record.

The benchmark cannot be used as a VRA safe harbor. Federally required remedies are judged against full legal compliance, not proximity to a race-excluded map.

What The Real Evidence Shows

The evidence program has become more credible because it reports failures.

- Two clean-source Rhode Island executions produced identical canonical assignments.
- Real Rust and GerryChain ensembles were archived for Rhode Island, Iowa, and North Carolina.
- Rhode Island supports a stable low cut-fraction result.
- Iowa and North Carolina lack enough effective samples for extreme-tail claims.
- Rust and GerryChain distributions differ materially in Iowa and North Carolina.
- Wisconsin was excluded because a benchmark district was disconnected.

These are not marketing defects. They show that the evidence system can reject its own preferred story.

The Huntington–Hill North Star

Huntington–Hill ended operational dispute after Congress selected the method. Redistricting is harder because it creates geographic regions rather than allocating seats among fixed States.

The closest equivalent is an **Exact Canonical Benchmark**:

1. Congress enacts the feasible-set axioms.
2. An exact solver minimizes maximum population deviation.
3. It then minimizes total population deviation.
4. It then minimizes geographic boundary cost.
5. A canonical GEOID assignment vector breaks all remaining ties.
6. A proof certificate establishes that no lexicographically better assignment exists under the identified enacted instance, model, and profile.

Given the enacted axioms and census inputs, the benchmark would be unique and mathematically certified. Debate remains over the axioms and lawful modifications—not over what the mathematics produced.

Gold-standard statement:

“For this enacted instance, model, and profile, this is the unique assignment selected by the enacted lexicographic objective. The attached certificate proves feasibility, model-relative optimality, and canonical tie-breaking.”

The Ask

Do not wait for exact optimization before adopting transparency. Adopt the evidence standard now.

Do not stop at a heuristic benchmark. Fund and test the Exact Canonical Benchmark until it can supply Huntington–Hill-like operational finality.

Claim Boundary

This brief describes an internal candidate standard. It is not enacted law, external peer review, block-level implementation, or proof that a BISECT plan is fair. The exact benchmark is a research and engineering goal.

Why We Are Not There Yet

Exact connected redistricting is computationally difficult. The current METIS benchmark is heuristic. Bounded exact split and recursive-tree certificates, hostile verifier corpora, an independent E0 verifier, OPB proof requests, and a connected Rhode Island block RCTX are now implemented. They are foundations, not a complete national exact solver.

The North Star requires:

- scalable block-level exact formulations;
- compact proof-loggable connectivity constraints;
- lower bounds matching the incumbent objective;
- canonical tie-optimality proofs;
- RoundingSat proof generation and independent VeriPB verification; and
- demonstrated performance from small States to California and Texas.

Adoption Path

Actor	Near-term use
Congress	Enact the benchmark, disclosure, governance, and exact-standard research program.
Commissions	Use the benchmark as a public comparison and explain every departure.
Courts	Use packages and audits as evidence; retain legal judgment.
States	Precommit criteria, publish alternatives, and fund community participation.
Civic groups	Submit communities and plans; challenge data, procedure, and substantive harms.
Researchers	Reproduce the evidence and attack the exact-certificate design.